PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T30 Classified

Cross-session • chapter-organised candidate

23 classified items • 70 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Unit 2 • Chapter OL-T30 • 23 items • 70 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Functions

Unit 2 • 70 marks • 23 item(s)

June 2025 | 4MA1-1H Q16 | 1 marks
Evaluate a function at a numerical input

June 2025 | 4MA1-1H Q16 | 3 marks
Solve a composite function inequality

June 2024 | 4MA1-1H Q15 | 4 marks
Exclude a denominator zero from the domain

November 2023 | 4MA1-2H Q19 | 1 marks
Exclude a denominator zero from the domain

November 2023 | 4MA1-2H Q19 | 2 marks
Evaluate a numerical composite function

November 2023 | 4MA1-2H Q19 | 4 marks
Inverse of a restricted quadratic function

June 2023 | 4MA1-1H Q17 | 4 marks
Exclude a denominator zero from the domain

November 2024 | 4MA1-1H Q25 | 4 marks
Inverse of a restricted quadratic function

June 2022 | 4MA1-1H Q14 | 1 marks
Evaluate a function at a numerical input

June 2022 | 4MA1-1H Q14 | 3 marks
Inverse of a linear or fractional function

January 2022 | 4MA1-2H Q25 | 4 marks
Inverse of a restricted quadratic function

January 2022 | 4MA1-2H Q25 | 1 marks
Domain and range swap under inversion

January 2020 | 4MA1-1H Q21 | 5 marks
Inverse of a composite-defined function

January 2021 | 4MA1-1H Q22 | 3 marks
Inverse of a restricted quadratic function

June 2021 | 4MA1-1H Q24 | 2 marks
Evaluate a numerical composite function

... 8 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

16 $f(x) = 9 - \sqrt{x}$ where $x \geqslant 0$

$g(x) = 4x^2$

(a) Find f(9)

....

(1)

WORKING SPACE

16 $f(x) = 9 - \sqrt{x}$ where $x \geqslant 0$

$g(x) = 4x^2$

(b) Solve $fg(x) < 0$

Show clear algebraic working.

....

(3)

WORKING SPACE

Question 07

4MA1-1H Q15

MODULAR UNIT 2

4 marks

15 The function f is defined as

$$f : x \mapsto \frac{3x + 1}{x - 2}$$

(a) State the value of x that cannot be included in any domain of the function f

....

(1)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

(3)

WORKING SPACE

Question 31

4MA1-2H Q19 (a)

MODULAR UNIT 2

1 marks

19 The functions f and g are such that

$$f : x \mapsto 5x + 7$$
$$g : x \mapsto \frac{5}{2x - 9}$$

(a) State which value of x cannot be included in any domain of g

....

(1)

WORKING SPACE

Question 32

4MA1-2H Q19 (b)

MODULAR UNIT 2
2 marks

19 The functions f and g are such that

$$\mathrm{f}: x \mapsto 5x + 7$$

$$\mathrm{g}: x \mapsto \frac{5}{2x - 9}$$

(b) Find fg(4)

....

(2)

WORKING SPACE

Question 33

4MA1-2H Q19 (c)

MODULAR UNIT 2
4 marks

The function h is such that

$$h : x \mapsto 3x^2 - 12x + 8 \quad \text{where } x > 2$$

(c) Express the inverse function h^{-1} in the form $h^{-1} : x \mapsto \ldots$

$h^{-1} : x \mapsto$ ....

(4)

WORKING SPACE

Question 14

4MA1-1H Q17

MODULAR UNIT 2
4 marks

17 The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \qquad x \geqslant 0$$

(a) What value of x must be excluded from any domain of g?

....
(1)

(b) Solve gh(x) = 1

....
(3)

WORKING SPACE

Question 17

4MA1-1H Q25 M01

MODULAR UNIT 2
4 marks

25 The function f is such that $f(x) = 2x^2 - 24x + 7$ where $x \geqslant 6$

Find the inverse function $f^{-1}(x)$

$f^{-1}(x) =$ ....

WORKING SPACE

Question 13

4MA1-1H Q14 Q14(a)

MODULAR UNIT 2
1 marks

14 The function f is defined as

$$f : x \mapsto \frac{2x}{x-6} \qquad x \neq 6$$

(a) Find f(10)

....

(1)

WORKING SPACE

Question 14

4MA1-1H Q14 Q14(b)

MODULAR UNIT 2

3 marks

14 The function f is defined as

$$f : x \mapsto \frac{2x}{x-6} \qquad x \neq 6$$

(b) Express the inverse function f^{-1} in the form $f^{-1} : x \mapsto \ldots$

$f^{-1} : x \mapsto$ ....

(3)

WORKING SPACE

Question 36

4MA1-2H Q25 Q25(a)

MODULAR UNIT 2
4 marks

25 The function g is defined as

$$\mathrm{g}: x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x : x \geqslant 3\}$$

(a) Express the inverse function g^{-1} in the form $\mathrm{g}^{-1} : x \mapsto \ldots$

$\mathrm{g}^{-1} : x \mapsto$ ....

(4)

WORKING SPACE

Question 37

4MA1-2H Q25 Q25(b)

MODULAR UNIT 2
1 marks

(b) State the domain of g^{-1}

....
(1)

WORKING SPACE

January 2020 | 4MA1-1H | Question 21 | 5 marks

Chapter OL-T30: Functions

21 The functions f and g are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function h is such that $h(x) = fg(x)$ for $x \geqslant -2$

Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \ldots$

$h^{-1}(x) = $ ....

YOUR WORKING

22 The function f is such that $f(x) = x^2 - 8x + 5$ where $x \leqslant 4$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

YOUR WORKING

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \qquad \text{where } x \geqslant 1$$

$$g(x) = 7x - 6$$

(a) Find fg(2)

....

(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

(4)

YOUR WORKING

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \qquad \text{where } x \geqslant 1$$

$$g(x) = 7x - 6$$

(a) Find fg(2)

....

(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

(4)

YOUR WORKING

15 The functions f and g are such that

$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x + 1}$$

(a) State the value of x that cannot be included in any domain of g

....

(1)

(b) Find gf(x)

Simplify your answer.

gf(x) = ...

(2)

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \ldots$

$g^{-1}(x)$ = ...

(3)

YOUR WORKING

15 The functions f and g are such that

$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x+1}$$

(a) State the value of x that cannot be included in any domain of g

....
(1)

(b) Find gf(x)
Simplify your answer.

gf(x) = ...
(2)

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = ...$

$g^{-1}(x)$ = ...
(3)

YOUR WORKING

15 The functions f and g are such that

$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x+1}$$

(a) State the value of x that cannot be included in any domain of g

....
(1)

(b) Find $gf(x)$
Simplify your answer.

$gf(x) =$ ....
(2)

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \ldots$

$g^{-1}(x) =$ ....
(3)

YOUR WORKING

May-Nov 2020 | 2H | Question 21 | 7 marks

Chapter OL-T30: Functions

21 The function f is such that $f(x) = 5 + 6x - x^2$ for $x \leqslant 3$

(a) Express $5 + 6x - x^2$ in the form $p - (x - q)^2$ where p and q are constants.

....

(2)

(b) Using your answer to part (a), find the range of values of x for which $f^{-1}(x)$ is positive.

....

(5)

(Total for Question 21 is 7 marks)

YOUR WORKING

18 The function f is such that $f(x) = \frac{k}{x}$ where $x \neq 0$ and k is an integer.

(a) Express the inverse function f^{-1} in the form $f^{-1}(x) = ...$

$f^{-1}(x) =$ ....

(1)

The function g is such that $g(x) = 2 - 3x^4$ where $x \neq 0$

The function h is such that $h(x) = \frac{3x}{2 - x}$ where $x \neq 2$

(b) (i) Find g(−2)

....

(1)

(ii) Express the composite function hg in the form $hg(x) = ...$
Give your answer in its simplest form.

$hg(x) =$ ....

(2)

(Total for Question 18 is 4 marks)

YOUR WORKING

January 2023 | 2H | Question 25 | 4 marks

Chapter OL-T30: Functions

25 The function f is such that $f(x) = 3x^2 - 12x + 7$ where $x \leqslant 2$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ ....

(Total for Question 25 is 4 marks)

YOUR WORKING

19 The functions f and g are such that

$$f(x) = 3x - 2 \qquad\qquad g(x) = \frac{x}{2x-1}$$

(a) Find g(3)

....

(1)

(b) Find gf(x)

Give your answer in its simplest form.

gf(x) = ...

(2)

(Total for Question 19 is 3 marks)

YOUR WORKING